New Energy Grid

⚡ FIELD INSTALLATION & COMMISSIONING MANUAL

GEMS Installer Manual

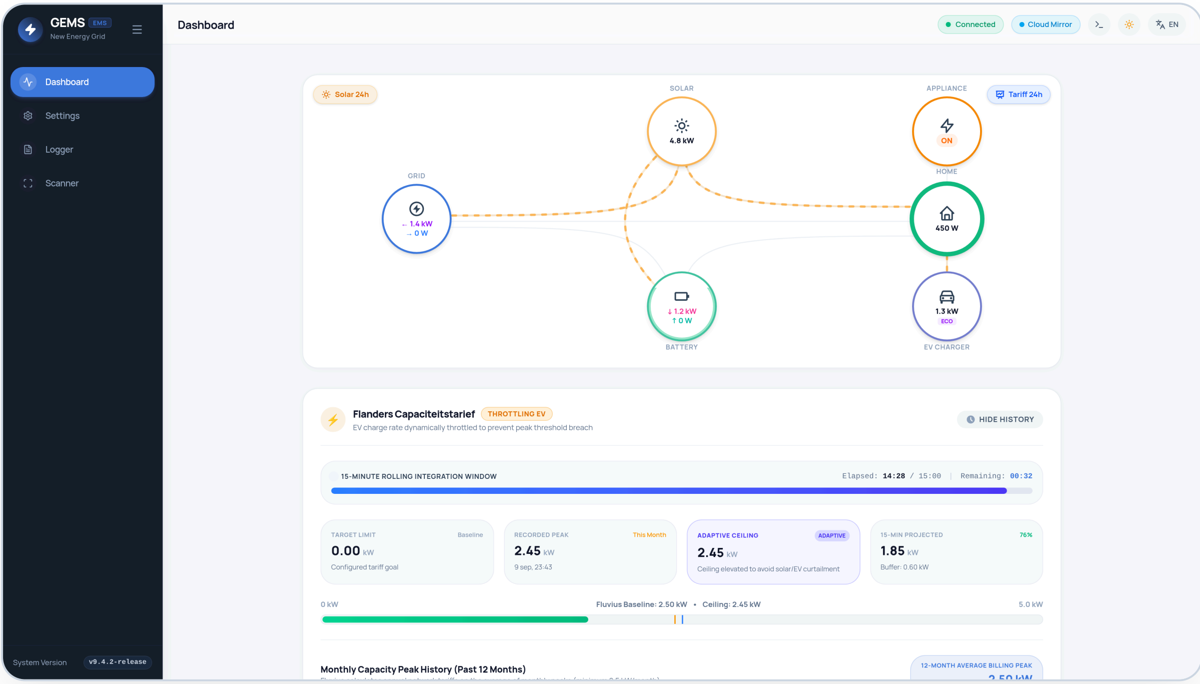
Hardware Setup, Network Discovery, Device Integration, and Optimization Strategy Configuration

SYSTEM FIRMWARE / BUILD
v9.6.1-release (Official Tag)

TARGET HARDWARE ARCHITECTURE
Raspberry Pi 5 (ARM64) + M.2 NVMe HAT

TARGET AUDIENCE
Electricians, Solar & ESS Installers, Integrators

REVISION DATE
September 2026



GEMS Energy Management System (GRID-EMS)   Confidential Field Reference • Documentation v9.6.1

TABLE OF CONTENTS

1. System Architecture & Hardware BOM

GEMS (Grid Energy Management System) is an autonomous, lightweight, fully UI-driven energy manager developed in Go and Vue 3. It runs directly on the local residential network on a Raspberry Pi without requiring cloud subscription dependencies. To eliminate SD card wear and ensure decades of reliable operation, GEMS runs from an M.2 NVMe SSD using SQLite configured in Write-Ahead Logging (WAL) mode.

Reference Hardware Bill of Materials (BOM)

Component	Specification	Installer Notes
Host Computer	Raspberry Pi 5 (4GB or 8GB RAM, Broadcom BCM2712)	Provides hardware PCIe 2.0 interface. CPU usage stays < 5% during full polling.
Internal Storage	Patriot P300 128GB M.2 PCIe Gen 3 ×4 NVMe SSD	Low power consumption, durable TLC flash. Write speed up to 600 MB/s.
M.2 PCIe HAT	Raspberry Pi 5 M.2 HAT+ (or db-tronic Bottom PCIe Base)	Connects to the 16-pin FPC PCIe ribbon connector. Ensure gold pins face correctly.
Power Supply	Official Raspberry Pi 27W USB-C PD (5.1V / 5.0A)	Mandatory to power both the Pi 5 and the NVMe SSD without brownouts.
Cooling Enclosure	Aluminium armor metal case with PWM Active Cooler	Ensures sustained 24/7 operating temperatures below 48°C in electrical cabinets.
Serial Interface	FTDI USB-to-RS485 Isolated Converter or P1 RJ12 cable	Connects to DSMR P1 digital meters or Modbus RTU serial buses (A/B lines).

2. NVMe Bootloader & EEPROM Setup

Brand new Raspberry Pi 5 boards ship with a factory EEPROM configured to boot solely from MicroSD (`BOOT_ORDER=0xf41`). When booting with the Patriot P300 NVMe SSD, the PCIe bus must be enabled via the bootloader.

🌟 **3-Minute Quick Fix (Using Raspberry Pi Imager)**

1. Insert any MicroSD card into your laptop.
2. Open **Raspberry Pi Imager** → Choose Device: **Raspberry Pi 5** → Choose OS: **Misc utility images** → **Bootloader** → **NVMe/PCIe Boot**.
3. Choose the SD card, click **Write**, and insert into the Pi 5.
4. Power on with the official 27W power supply. Within 5 seconds, the green ACT LED will flash rapidly and the HDMI screen will turn **solid green**.
5. Unplug power and discard/remove the MicroSD card. The EEPROM is now permanently set to `BOOT_ORDER=0xf461` and `PCIE_PROBE=1`.

3. OS Deployment & System Services

GEMS provides two official deployment methods for installers:

Option A: Pre-built Custom Raspberry Pi OS Image (Recommended)

Download `gems-os-image.img.xz` from the GitHub release. Flash it directly to the Patriot P300 SSD via an external USB NVMe enclosure using BalenaEtcher or Raspberry Pi Imager. The pre-built image includes:

- **Nginx Proxy:** Routes port 80/443 to the GEMS backend on port 8080 with WebSocket and SSE support.
- **Kiosk Mode:** Minimal Wayland compositor (Wayfire) displaying Chromium in fullscreen pointing to `http://localhost`.
- **Cockpit Web Console:** Available on port `9090` for browser-based root terminal access.
- **Raspberry Pi Connect:** Pre-installed daemon for secure browser-based remote screen viewing.

Option B: Debian Package (.deb) on Existing OS

```
# Download and install the latest ARM64 Debian package
sudo dpkg -i gems_9.6.1_arm64.deb
sudo apt-get install -f

# Verify service status
sudo systemctl status gems.service
```

The package automatically provisions a dedicated system user (`gems`), installs binaries to `/opt/gems`, and enables the systemd service.

4. Remote Access & Diagnostics Console

Technicians can manage the installation without keyboard or monitor:

- **Web UI:** `http://ems.local` or `http://<IP-ADDRESS>` (Default port 80 or 8080).
- **Cockpit Web Terminal:** `https://<IP-ADDRESS>:9090` (Default User: `admin`, Password: `manufacturer`). Offers web terminal, storage SMART metrics, system reboot, and networking.
- **Remote Screen Sharing:** Run `rpi-connect signin` in Cockpit to link the Pi to your Raspberry Pi ID for one-click remote desktop view.

5. Subnet Hardware Discovery (Scanner)

Before configuring devices, use the built-in **Scanner** in the sidebar. The scanner performs a parallel ARP sweep and TCP port scan across the local /24 subnet, matching MAC addresses against known solar and meter vendors (Huawei, SMA, Eastron, Shelly, Alfen, Easee).

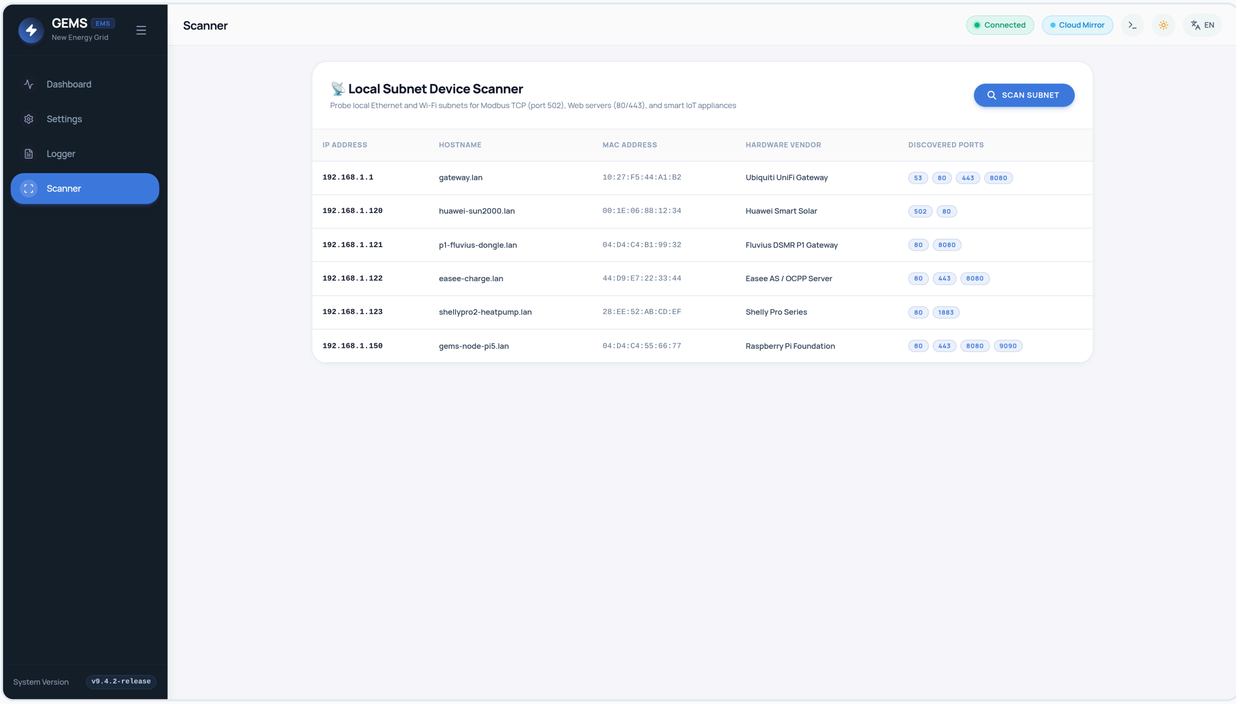


Figure 5.1: GEMS Network Scanner identifying inverter dongles, smart meters, and EV chargers with open ports.

Scanner Usage Steps

1. Click **Scanner** in the primary left navigation.
2. Click the blue **Scan Subnet** button.
3. Review the discovered hosts: note the detected IP, open ports (e.g. 502 for Modbus TCP, 80 for REST), and vendor tags.

6. Adding & Configuring Devices (UI-Driven)

GEMS includes **90 native hardware templates**. Devices are added exclusively through the web interface with zero YAML file editing.

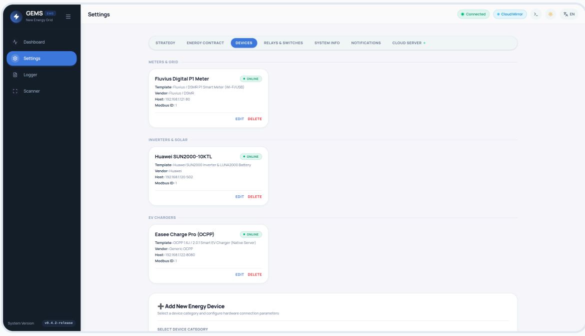


Figure 6.1: Active Devices list with live health tags.

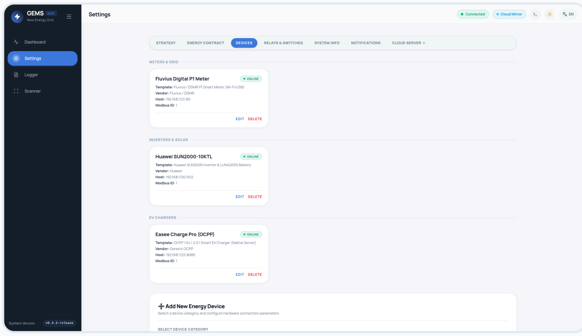


Figure 6.2: Device category picker wizard.

Step-by-Step Onboarding Procedures

A. Solar & Hybrid Inverters (Modbus TCP)

Supported brands include Huawei SUN2000, SMA Sunny Boy / Tripower, Solis 5G/S6, Fronius Gen24, SolarEdge, Sungrow, GoodWe, and Victron GX.

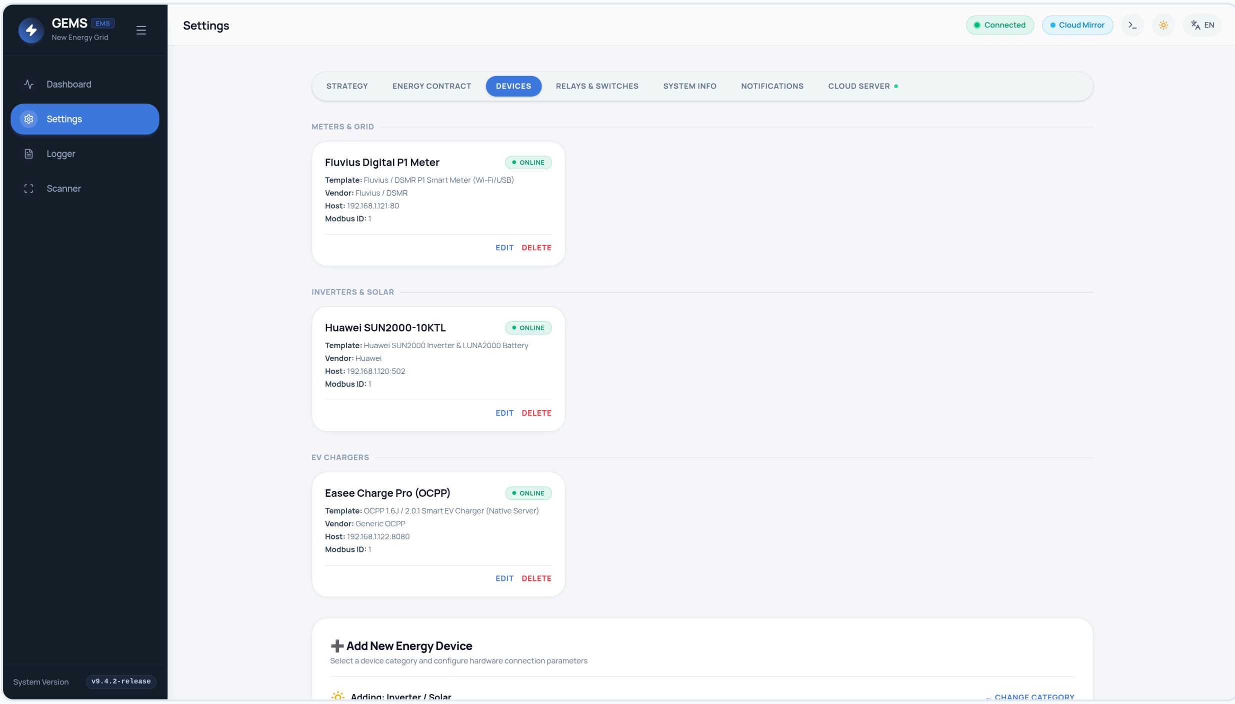


Figure 6.3: Solar Inverter Modbus TCP configuration modal.

Field	Typical Value	Description
`Template`	Huawei SUN2000 / SMA / Solis	Selects the pre-mapped SunSpec/Modbus register map.
`Host IP`	`192.168.1.120`	IP address of the Wi-Fi/Ethernet dongle (e.g. Huawei SDongleA-05).
`Port`	`502` or `1502`	Modbus TCP port. Huawei uses 502; some third-party bridges use 1502.
`Modbus Slave ID`	`1` (Huawei/SMA), `126` (SolarEdge)	Slave ID configured on the inverter internal bus.
`Rated Power (kW)`	e.g. `10.0`	Maximum rated AC output power of the inverter.
`Has Battery`	Checked (if LUNA2000/BYD present)	Enables battery telemetry, SOC tracking, and charge/discharge control.

B. Grid Smart Meters (P1 / Modbus)

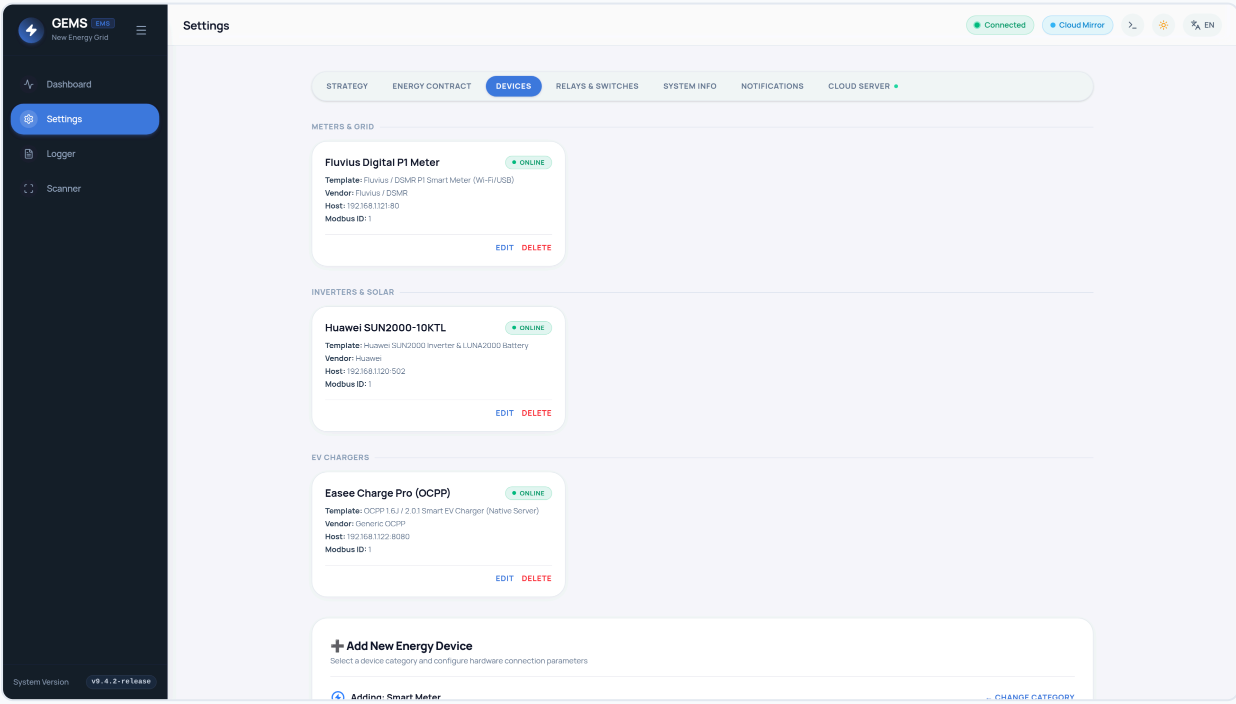


Figure 6.4: Grid Smart Meter configuration modal (P1 USB, TCP Bridge, SDM630, Shelly 3EM).

- **Fluvius / DSMR P1 Serial:** Set host to `/dev/ttyUSB0`, Baud rate `115200` (DSMR 5.0).
- **P1 Wi-Fi Dongle (HomeWizard / TCP Bridge):** Set host to dongle IP (e.g. `192.168.1.121`), port `80` or `8080`.
- **Eastron SDM630 (Modbus):** Connect RS485 A/B lines to USB converter, select template `Eastron SDM630 3-Phase`, Slave ID `1`.

C. EV Charging Stations (Modbus TCP & Native OCPP)

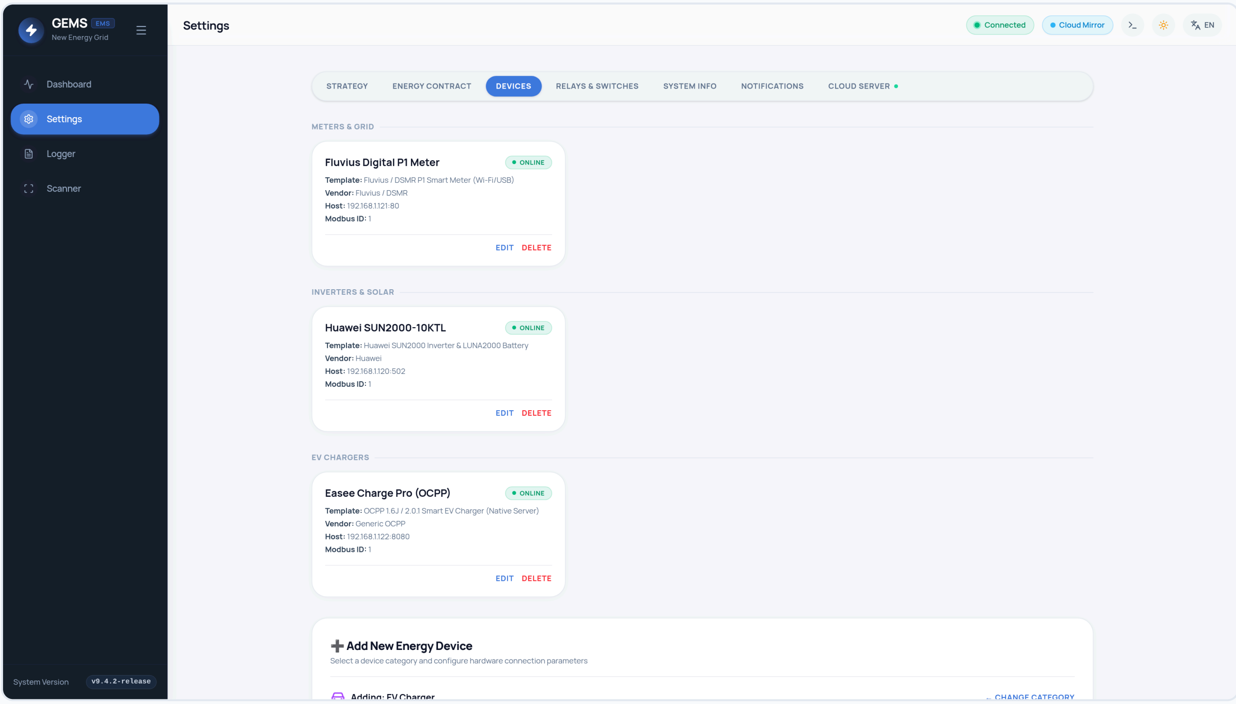


Figure 6.5: EV Charger configuration modal supporting preferred Modbus TCP direct integration and native OCPP server.

⚡ Modbus TCP Integration (Recommended & Preferred)

Modbus TCP is the preferred connection method for EV chargers. Connecting via Modbus TCP (Port 502) offers critical advantages over OCPP:

- **Sub-Second Setpoint Control:** Direct register writes allow instantaneous charge throttling (6A to 32A), essential for reactive Flanders capacity peak shaving and real-time solar curtailment tracking.
- **Simultaneous CPO / Billing Coexistence:** If the charger is already connected to an external Charge Point Operator (CPO) or employer billing reimbursement backend (e.g. E-Flux, Road, Optimile, Easee Cloud) via OCPP, GEMS controls the charger locally over Modbus TCP *without disconnecting or interfering with the cloud billing service.*
- **Zero WebSocket Latency:** Deterministic master-slave fieldbus polling on the local LAN with zero external broker dependencies.

Setup Steps: In your EV charger's web interface or installer tool, enable *Modbus TCP* (default port `502`) and note the Unit/Slave ID (typically `1`). In GEMS, click **Add Device → EV Charger**, choose your hardware template (e.g. *Alfen Eve*, *Keba KeContact P30*, *Mennekes Amtron*, *Webasto Live/Next*, *ABB Terra AC*, *Raedian Gemini/Neo*, *Phoenix Contact Charx*, *Veton*, *SolarEdge Home EVSE*, or *Generic Modbus TCP Charger*), enter the charger's local IP address, Port `502`, and Slave ID.

🔌 Native OCPP 1.6-J / 2.0.1 WebSocket Server (Alternative)

For chargers without Modbus TCP capability or installations without third-party CPO billing backends, GEMS runs an embedded WebSocket OCPP server on port **8887**:

```
Central System URL: ws://<GEMS-IP>:8887/<ChargePointID>
Example: ws://192.168.1.150:8887/CP001
```

In GEMS: **Add Device** → **EV Charger** → **Template: OCPP 1.6J / 2.0.1 Smart EV Charger**. Enter the matching `ChargePointID`. Once the charger establishes the WebSocket handshake, status turns to **● Online**.

7. Grid Connection & Multi-Phase Setup

Navigate to **Settings** → **Strategy** → **Grid Connection & Limits** to specify electrical supply constraints:

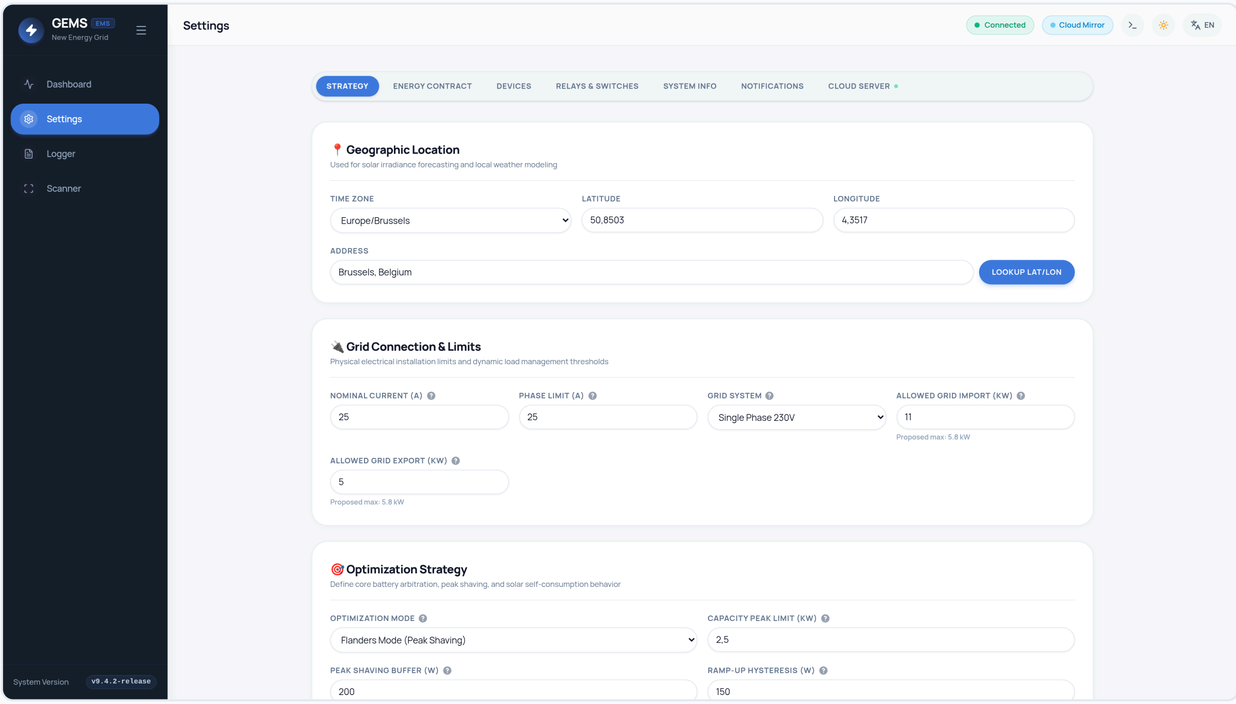


Figure 7.1: Grid connection parameters, phase limit amps, and maximum import/export allowance.

Parameter	Format	Field Explanation
`Grid System`	Dropdown	`Single Phase 230V`, `3-Phase 400V+N`, or `3-Phase 230V Delta (No Neutral)`.
`Nominal Current (A)`	Float (Amps)	Main circuit breaker rating (e.g. `25A`, `32A`, `40A`, `63A`).
`Phase Limit (A)`	Float (Amps)	Hard safety ceiling per phase. The EMS throttles EV charging if any phase exceeds this limit.
`Allowed Grid Import (kW)`	Float (kW)	Maximum instantaneous import power allowed by the connection.
`Allowed Grid Export (kW)`	Float (kW)	Maximum grid feed-in allowed by DNO contract. Set to `0.0` for strict zero-export.

8. Dynamic Contracts & 24h Tariff Curve

Under **Settings → Energy Contract**, configure the customer's utility contract. GEMS fetches hourly EPEX Spot Day-Ahead prices daily at 13:00 CET and calculates true retail prices including supplier markup, DNO distribution tariffs, and taxes.

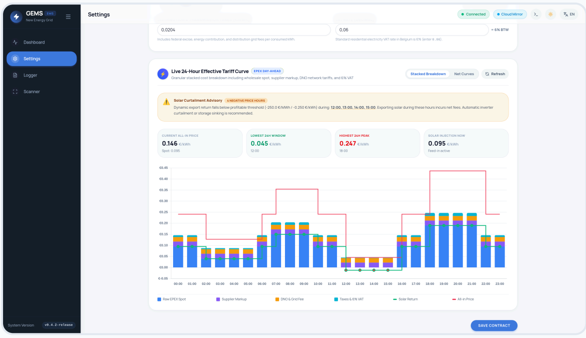


Figure 8.1: Belgian supplier contract presets and tax formulas.

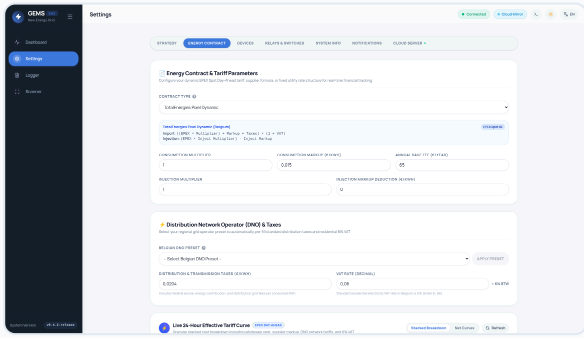


Figure 8.2: 24-Hour effective retail tariff breakdown visualizer.

Supported Contract Models & Supplier Presets

- **Dynamic EPEX Spot:** TotalEnergies Pixel Dynamic, Mega Smart/Cosy Dynamic, Bolt Dynamisch, Engie Dynamic / Flextime, Luminus, Eneco, Frank Energie.
- **DNO Network Tariff Presets:** Fluvius (Flanders), SIBELGA (Brussels), ORES / RESA (Wallonia).
- **Fixed & Dual-Tariff (Peak / Off-Peak):** Configurable peak kWh, off-peak kWh, and fixed injection tariff.
- **Taxes & VAT:** Configurable federal excise duty (default €0.0204/kWh) and Belgian 6% residential VAT.

9. Optimization Strategy & Flemish GSC Protection

Navigate to **Settings → Strategy** to configure peak shaving, active inverter curtailment, and green certificate value protection.

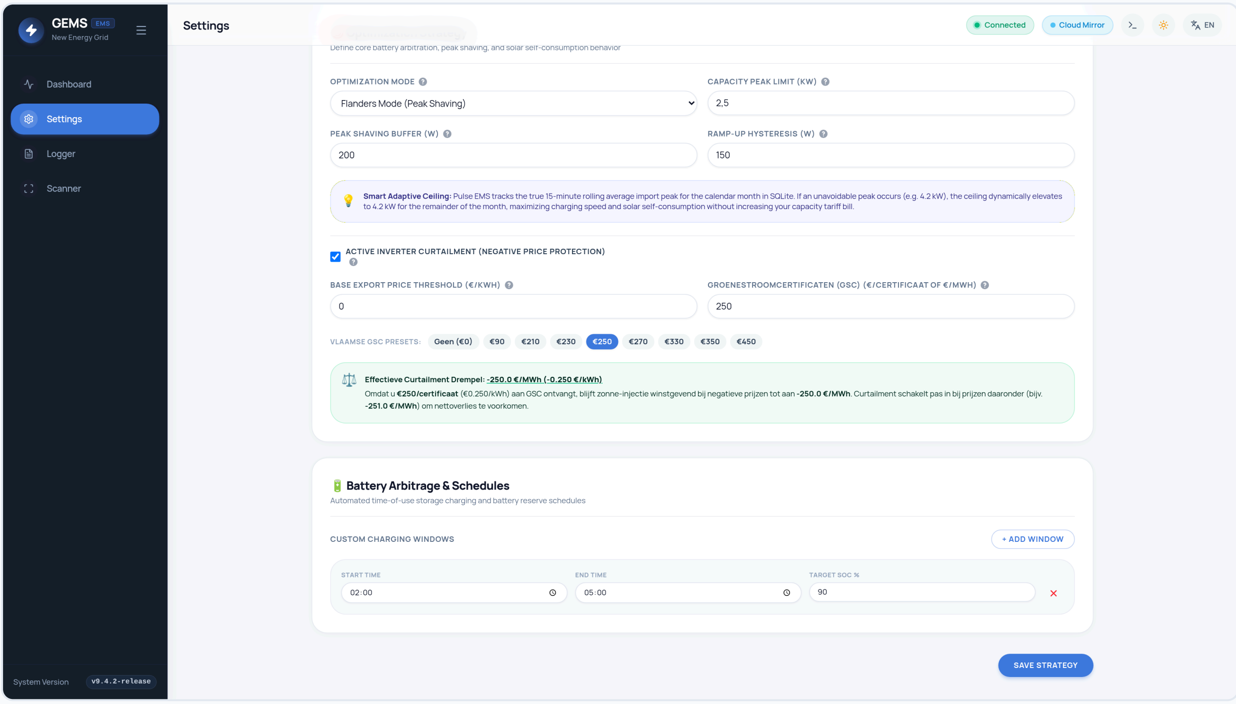


Figure 9.1: Flanders Peak Shaving, Smart Adaptive Ceiling, and Flemish GSC Value presets.

Flanders Capacity Tariff (Capaciteitstarief)

In Flanders, grid distribution fees are calculated based on the homeowner's monthly peak 15-minute average import power (minimum baseline 2.5 kW).

- `capacity_peak_limit_kw`: Hard billing ceiling (default **2.50 kW**).
- `peak_shaving_buffer_w`: Safety margin (default **200 W**) to account for appliance startup spikes.
- `peak_shaving_rampup_w`: Re-acceleration hysteresis rate (default **150 W**).
- **Smart Adaptive Ceiling:** If a unavoidable household peak occurs (e.g. 4.2 kW from oven + heat pump), GEMS dynamically adapts its operational ceiling to 4.2 kW for the remainder of that calendar month. This allows maximum EV charging speed and solar self-consumption without raising the monthly electricity bill!

Active Inverter Curtailment & Groenestroomcertificaten (GSC) Value Protection

During negative spot price hours, exporting solar energy can result in financial penalties. However, legacy Flemish PV systems receive guaranteed **Groenestroomcertificaten (GSC)** (e.g. €250, €350, or €450 per 1,000 kWh).

💶 How GSC Protection Prevents Financial Loss

If a customer receives **€250/certificaat (€0.250/kWh)**, solar injection remains profitable even if the market injection price drops to **-€0.20/kWh (-200 €/MWh)** because the homeowner still nets + €0.05/kWh!

```
Effective Curtailment Threshold = min_profitable_export_price - (gsc_value_eur_mwh / 1000.0)
Example: €0.00 - (€250 / 1000) = -0.250 €/kWh (-250.0 €/MWh)
```

GEMS will **NOT** curtail the inverter during moderately negative hours. Curtailment only engages if the negative price drops below -250.0 €/MWh (e.g. at -251.0 €/MWh), maximizing the customer's total subsidy earnings!

10. Smart Relays & Heat Pump SG-Ready Automation

GEMS integrates Shelly Pro 2 and Shelly Plus 1PM contactors to automate thermal loads (heat pumps, hot water boilers) based on real-time solar surplus or negative spot pricing.

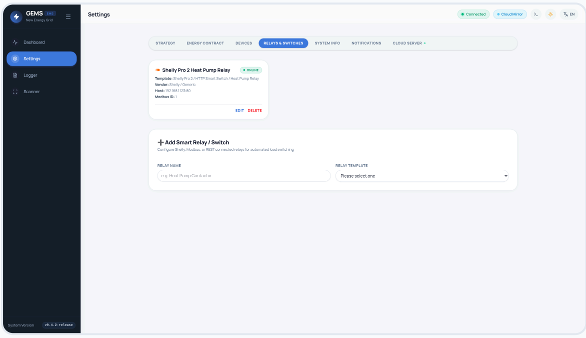


Figure 10.1: Configured relays and real-time ON/OFF state.

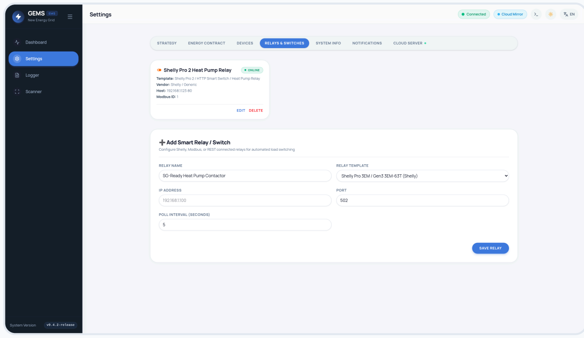


Figure 10.2: Excess solar and negative spot price trigger rules.

Typical Automation Trigger Scenarios

Rule Type	Trigger Condition	Action	Hysteresis	Application
`excess_solar`	Surplus > 1,200 W	Turn ON	10 Minutes	SG-Ready Heat Pump thermal boost to elevate buffer tank temperature.
`negative_spot_price`	Tariff ≤ €0.02/kWh	Turn ON	15 Minutes	Domestic hot water boiler electrical resistance element heating.

11. Alerts, Webhooks & Automated Reports

Configure external monitoring under **Settings → Notifications**:

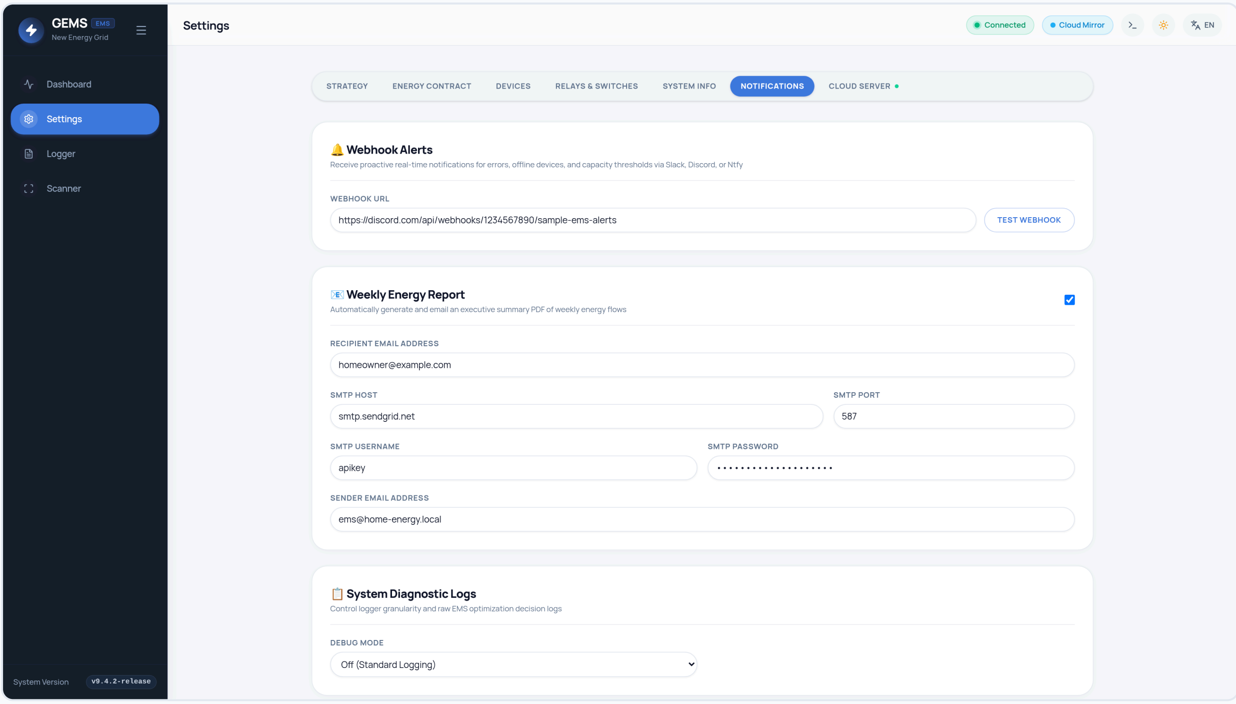


Figure 11.1: Discord/Slack webhook URLs, email SMTP configuration, and weekly report recipient.

- **Discord / Slack / Teams Webhooks:** Real-time alerts on consecutive device polling failures, Modbus communication timeouts, and monthly capacity peak ceiling breaches.
- **Weekly Homeowner Email:** Automatic delivery of a weekly performance summary with total solar generation, self-consumption percentage, and peak grid import.

12. System Diagnostics, Logger & OTA Updates

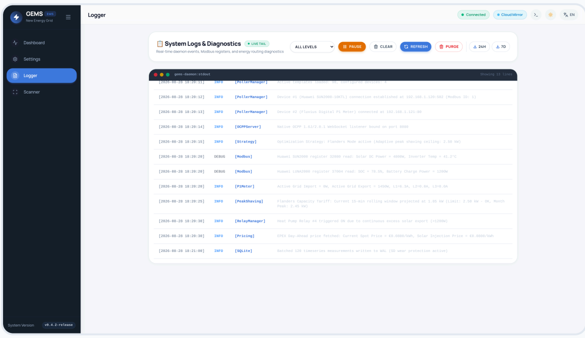


Figure 12.1: Real-time system diagnostics logger.

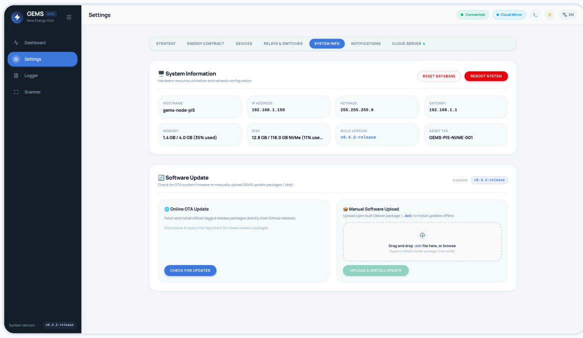


Figure 12.2: NVMe SSD health, build version, and OTA update tool.

Field Commissioning Checklist

- [ ] Inverter Modbus TCP responding with valid solar wattage on the Dashboard.
- [ ] Smart meter (P1 or SDM630) reporting import/export power consistent with utility meter.

- [ ] EV Charger connected via Modbus TCP (preferred) or native OCPP WebSocket (`ws://<IP>:8887/<CP-ID>`) and reporting live current setpoint.
- [ ] Flanders Capacity Tariff limit set according to customer agreement (default 2.50 kW).
- [ ] GSC certificate value configured if the installation is eligible for Flemish green certificates.
- [ ] Energy contract provider and distribution DNO selected under Energy Contract tab.
- [ ] Weekly report email address tested and saved.